

DHEERAJ KUMAR

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[LinkedIn](#) — [GitHub](#) — [LeetCode](#)

Education

National Institute of Technology (NIT), Trichy

2022 – 2026

B.Tech in Civil Engineering | Minor in Computer Science CGPA: 7.64

Technical Skills

- **Languages:** Python (OOP, data structures, functions), SQL, C/C++
- **AI / LLMs:** LangChain, LangGraph, RAG Pipelines, Prompt Engineering, LLM APIs, Embeddings, Semantic Search, TensorFlow
- **NLP:** Text Classification, Summarization, Question Answering
- **Machine Learning:** Scikit-learn, Feature Engineering, Cross-Validation, Evaluation Metrics (Precision, Recall, F1)
- **Data & Databases:** Pandas, NumPy, Jupyter Notebook, PostgreSQL, MongoDB, MySQL
- **Tools:** Git, Docker, FastAPI, AWS Bedrock

Experience

Data Science Research Intern — NIT Trichy

May 2025 – July 2025

- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain, MongoDB, and semantic search with LLMs for contextual corporate-report Q&A – directly covering document ingestion, embedding, and context retrieval.
- Implemented prompt engineering improvements and integrated AI features with REST APIs for internal applications, monitoring latency and accuracy.
- Conducted EDA on multi-million record banking datasets using Python, Pandas, and SQL; applied PCA and K-Means clustering to segment 100K+ customers, reducing manual analysis time by 40%.
- Preprocessed and cleaned large structured datasets (labeling, feature engineering) for downstream ML model training and evaluation.

Projects

AI-Powered Multi-Agent BI Platform [GitHub](#) | [Live](#)

June 2026

Python, FastAPI, LangGraph, LLM APIs, PostgreSQL, XGBoost, TensorFlow, Groq

- Engineered a 5-agent LangGraph workflow for automated data cleaning, analytics, forecasting (XGBoost, ARIMA), and LLM-driven executive report generation – integrated via FastAPI and PostgreSQL.
- Used LLM APIs for summarization and classification tasks within the agent pipeline, with prompt engineering to improve output quality and consistency.
- Reduced end-to-end business analysis time by 90%+ (from hours to under 30 seconds); monitored model latency and accuracy for each agent stage.

Layoff Risk Prediction [GitHub](#)

2025

Python, Scikit-learn, PCA, Pandas, Jupyter Notebook

- Built a supervised ML classification model to predict company layoff risk from structured HR and financial data using Scikit-learn.
- Applied PCA for dimensionality reduction and feature engineering to improve signal quality; evaluated with cross-validation, achieving 79% accuracy on held-out data.
- Documented the full ML pipeline in Jupyter Notebook with clean, modular Python code covering preprocessing, training, and evaluation.

Multi-Agent Travel AI [GitHub](#) | [Live](#)

May 2026

Python, LangGraph, LangChain, LLM APIs, PostgreSQL, Tavily, Docker

- Architected a sequential LangGraph multi-agent pipeline using LLM APIs for NLP-driven travel planning – covering flight search, hotel lookup, itinerary creation, and summarization.
- Integrated Tavily for semantic web search and PostgreSQL for stateful trip persistence; containerized with Docker and deployed on Render.
- Implemented rate limiting, user feedback collection, and downloadable Markdown/PDF outputs – following clean API integration and documentation practices.

Achievements & Certifications

- [Oracle AI Foundations Associate](#) & [Oracle Generative AI Professional](#) – certified in LLM fundamentals and generative AI application development.
- **Naukri Scholarship 2025:** College Rank #2, National Rank #614 among 18,000+ engineering students.
- **HCLTech Hackathon 2026:** Developed solutions for real-world data analytics challenges.
- **National Mathematics Olympiad (NMO):** AIR 513.